

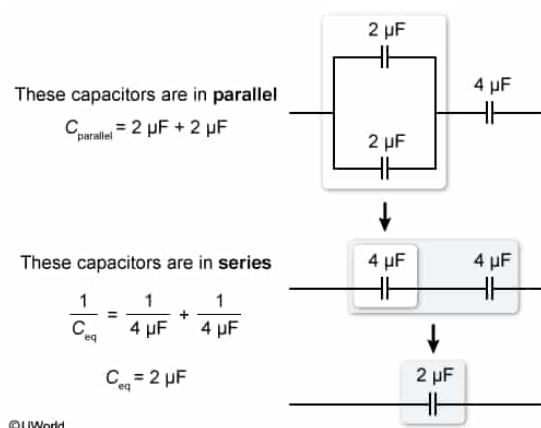
A single 4-μF capacitor is added in series with the set of parallel capacitors shown above. What is the ratio of the new equivalent capacitance to the original equivalent capacitance?

- ☐ A. 1:5 [8%]
- ✓ ☒ B. 1:2 [46%]
- ☐ C. 2:1 [32%]
- ☐ D. 5:1 [11%]

Correct

46% answered correctly

Explanation:



The 2-μF capacitors are connected in **parallel** because they are both connected to the same pair of wires. For capacitors connected in **parallel**, the equivalent circuit capacitance is the **sum of the individual capacitances**:

$$C_{\text{parallel}} = C_1 + C_2 + C_3 + \dots$$

Therefore, the equivalent capacitance of the original circuit is:

$$C_{\text{parallel}} = 2 \mu F + 2 \mu F = 4 \mu F$$

For capacitors in **series**, the inverse of the equivalent capacitance is the **sum of the inverses** of the individual capacitances:

$$\frac{1}{C_{\text{series}}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots$$

For capacitors in series, the circuit equivalent capacitance is always smaller than the smallest individual capacitance in the circuit.

When the 4- μF capacitor is added in series with the given capacitor circuit, the equivalent capacitance of the new circuit is:

$$\frac{1}{C_{\text{eq}}} = \frac{1}{4 \mu \text{ F}} + \frac{1}{4 \mu \text{ F}} = \frac{2}{4 \mu \text{ F}} = \frac{1}{2 \mu \text{ F}}$$
$$C_{\text{eq}} = 2 \mu \text{ F}$$

Therefore, the equivalent capacitance decreased to **2 from 4 μF** , giving a new-to-original ratio of **1:2**.

(Choices A and D) If the equations for the equivalent capacitance in series and in parallel were switched, the original equivalent capacitance would be 1 μF , and the new equivalent capacitance would be 5 μF . Therefore, the new-to-original ratio would be calculated incorrectly as 5:1.

(Choice C) If the 4- μF capacitor is placed in parallel with the other two capacitors, the new equivalent capacitance would be 4 μF + 2 μF + 2 μF = 8 μF . Therefore, the new-to-original ratio would be calculated incorrectly as 2:1.

Educational objective:

For capacitors in parallel, the equivalent capacitance is the sum of the individual capacitances: $C_{\text{parallel}} = C_1 + C_2 + C_3 \dots$. For capacitors in series, the inverse of the equivalent capacitance is the sum of the inverses of the individual capacitances:

$$\frac{1}{C_{\text{series}}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots$$

Time spent: 0 sec

Subject:
Physics

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